

Modeling and Forecasting VN-Index Volatility: A Financial Econometric Evaluation of GARCH-Family, HAR, Neural, and Hybrid Approaches

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Abstract—This paper studies one-step-ahead daily VN-Index volatility forecasting using CafeF OHLC and trading-activity data. The target is next-day squared percentage log return, and all primary forecasts are evaluated on a unified out-of-sample window of 726 common target dates from 2023-02-07 to 2025-12-31. The study compares historical and rolling variance baselines, EWMA/RiskMetrics, HAR-type models, GARCH(1,1), ARIMA-GARCH, validation-selected GARCH-family specifications, LSTM, hybrid neural forecasts, calibration, and forecast combinations. Descriptive diagnostics show negatively skewed, heavy-tailed returns, and ARCH-LM tests strongly reject homoskedasticity, motivating conditional-variance models. On the corrected common window, AdvGARCH-BestAsymmetric obtains QLIKE 1.0214, the primary validation-selected AdvGARCH-BestQLIKE obtains 1.0229, HAR-Parkinson obtains 1.0294, and GARCH(1,1) obtains 1.0529. The evidence supports a conservative class-level conclusion: selected GARCH-family specifications and HAR-Parkinson form the strongest group, while the implemented neural and hybrid approaches do not robustly outperform these econometric benchmarks in this modest daily-data setting. Risk diagnostics qualify the average-loss ranking. Under a common-Normal variance-to-VaR mapping, GARCH(1,1) has 15 one-percent violations out of 726 observations, against 7.26 expected, with Kupiec $p = 0.0116$. Thus good QLIKE does not by itself guarantee strict-tail calibration.

Index Terms—VN-Index, volatility forecasting, GARCH, EGARCH, LSTM, hybrid forecasting, QLIKE, VaR backtesting, range-based volatility.

I. INTRODUCTION

Volatility forecasting is a practical market-risk problem. For an equity index, a one-step-ahead conditional-variance forecast summarizes the expected scale of the next return shock and feeds into risk limits, margin decisions, portfolio monitoring, and Value-at-Risk (VaR). A model can look accurate on average while still being too low when volatility spikes, which is exactly when risk forecasts matter most.

The VN-Index is a useful Vietnamese equity-market case study. It is an emerging-market index with visible volatility clustering, sharp drawdowns, and changing trading conditions, but without the routine high-frequency realized-volatility data often used in developed-market volatility studies. Daily open, high, low, close, volume, and trading-value data therefore create a realistic empirical setting: forecasts must be made

from daily information, and latent volatility must be evaluated through imperfect observable proxies.

This paper is positioned as a reproducible empirical financial econometrics study, not as a new forecasting algorithm. The main question is whether daily VN-Index volatility can be forecast reliably using standard financial econometric models, and whether neural or hybrid alternatives provide robust gains once model selection, proxy uncertainty, and tail-risk diagnostics are considered. The main target is next-day squared percentage log return, and the primary comparison uses a leakage-safe common test window with identical dates for all primary models.

The contribution is fourfold:

- 1) A reproducible, leakage-safe out-of-sample comparison of daily VN-Index volatility forecasts on a unified 726-observation test window.
- 2) Econometric evidence on VN-Index return non-normality, volatility clustering, ARCH effects, and conditional-variance persistence.
- 3) A validation-based comparison of EWMA, HAR, GARCH-family, neural, hybrid, calibrated, and combined volatility forecasts.
- 4) Proxy-robustness, spike-underprediction, refit-robustness, and common-Normal VaR diagnostics that connect average forecast accuracy to practical risk interpretation.

The verified evidence supports a conservative conclusion. Selected GARCH-family specifications and HAR-Parkinson form the strongest group of daily VN-Index volatility forecasts under the unified common-window protocol. The results are economically consistent with persistent conditional variance and useful OHLC range information. The implemented neural and hybrid forecasts may contain complementary information, but they do not robustly dominate the strongest econometric benchmarks. Tail-risk diagnostics further show that good average QLIKE does not automatically imply adequate one-percent VaR coverage under a common-Normal mapping.

II. RELATED WORK

ARCH and GARCH models provide the classical foundation for conditional volatility forecasting. Engle introduced

ARCH to model time-varying conditional variance, and Bollerslev generalized the recursion to include lagged conditional variance [1], [2]. The empirical appeal of GARCH(1,1) is not only parsimony. Large-scale forecast comparisons have shown that it can be difficult to beat in practical volatility forecasting applications [3].

Mean dynamics are often modeled before conditional variance. The Box-Jenkins ARIMA framework captures serial dependence in univariate time series [4]. In financial returns, the conditional mean is usually small relative to the conditional variance, but an ARMA or ARIMA mean model can still be paired with GARCH to test whether residual variance forecasts improve after filtering predictable mean components.

EWMA/RiskMetrics is a standard industry benchmark for volatility and VaR forecasting because it provides a transparent exponentially weighted recursion with very few tuning choices [5]. Its inclusion matters empirically: if a sophisticated model does not clearly improve on EWMA under volatility-specific loss and tail diagnostics, the additional complexity is hard to justify. **EWMA is the minimum credible industry benchmark for this setting.**

HAR-type models provide another important benchmark class. The heterogeneous autoregressive realized-volatility model captures persistence at daily, weekly, and monthly horizons through a simple linear structure [6]. Even when high-frequency realized volatility is unavailable, HAR-style regressions using squared-return or OHLC range proxies are useful because they test whether multi-horizon volatility persistence explains daily variance better than naive rolling volatility.

GARCH-family extensions address stylized features that a symmetric GARCH(1,1) recursion may miss. EGARCH models allow asymmetric volatility responses and avoid some positivity restrictions [7]. GJR-type models capture sign-dependent shock effects [8]. APARCH and FIGARCH specifications allow richer power transformations and long-memory volatility dynamics [9], [10]. These extensions motivate the broad specification search evaluated in this paper.

Neural sequence models offer a different forecasting premise. LSTM networks were designed to learn long-range sequential dependencies through gated recurrent states [11]. In volatility forecasting, LSTMs can use lagged returns, volume, rolling volatility, and econometric forecasts as supervised features. Hybrid econometric-neural designs are therefore useful empirical candidates, but **their performance must be judged against strong volatility-specific baselines rather than weak naive alternatives.**

Forecast evaluation is central because latent volatility is not directly observed. Squared daily returns are available and simple, but they are noisy proxies for conditional variance. QLIKE is widely used because it remains attractive when forecast comparisons rely on imperfect volatility proxies and because it evaluates forecasts on the variance scale [12]. High-frequency realized volatility can provide richer targets when intraday data are available [13], but daily OHLC estimators are also useful. Parkinson, Garman-Klass, Rogers-Satchell,

TABLE I
CHRONOLOGICAL DATA SPLITS AND COMMON EVALUATION WINDOW.
THE COMMON-WINDOW COMPARISON ALIGNS ALL ORIGINAL FORECASTS
ON 726 TARGET DATES FROM 2023-02-07 TO 2025-12-31.

Dataset	Rows	Origin date range	Target date range
Prepared full	3989	2010-01-04 to 2025-12-31	2010-01-05 to 2025-12-31
Model-ready	3968	2010-02-01 to 2025-12-30	2010-02-02 to 2025-12-31
Train	2471	2010-02-01 to 2019-12-30	2010-02-02 to 2019-12-31
Validation	750	2020-01-02 to 2022-12-29	2020-01-03 to 2022-12-30
Test	745	2023-01-03 to 2025-12-30	2023-01-04 to 2025-12-31
Common test	726	2023-02-06 to 2025-12-30	2023-02-07 to 2025-12-31

and Yang-Zhang estimators exploit high, low, open, and close information to obtain alternative volatility proxies [14]–[17].

Statistical forecast comparisons require care when many models are examined. Diebold-Mariano tests compare predictive accuracy under a specified loss difference process [18]. Holm correction controls family-wise error rates across multiple tests [19]. The model confidence set (MCS) framework provides a complementary way to identify a set of models not statistically eliminated under a loss criterion [20]. VaR backtesting adds a risk-management view through unconditional and conditional coverage tests [21], [22]. **Multiplicity-aware inference is necessary before claiming single-model superiority.**

III. DATA AND ECONOMETRIC MOTIVATION

A. Daily VN-Index Data

The empirical sample uses CafeF VNINDEX daily observations from 2010 through 2025. The cleaned data include open, high, low, close, matched trading volume, and matched trading value. Table I reports the chronological splits. The model-ready sample contains 3968 rows from 2010-02-01 to 2025-12-30 after lagged-return, rolling-volatility, and next-day-target requirements are imposed.

The split is chronological. Training observations end in 2019, validation covers 2020–2022, and the test period begins in 2023. The authoritative primary leaderboard uses the common intersection of all primary prediction files: forecast origins from 2023-02-06 to 2025-12-30 and target dates from 2023-02-07 to 2025-12-31, with $N = 726$ for every primary model.

B. Return and Target Definitions

Let C_t denote the VN-Index close on trading day t . Percentage log return is

$$r_t = 100 \log \left(\frac{C_t}{C_{t-1}} \right), \quad (1)$$

and the daily squared-return volatility proxy is

$$v_t = r_t^2. \quad (2)$$

The supervised one-step target is

$$y_t = v_{t+1} = r_{t+1}^2. \quad (3)$$

Thus a forecast issued at origin date t uses information available through t to predict the variance proxy for target

TABLE II
DESCRIPTIVE STATISTICS FOR DAILY VN-INDEX PERCENTAGE LOG RETURNS.

Statistic	Value
Observations	3968
Mean	0.032738
Std. dev.	1.174558
Minimum	-6.910236
Maximum	6.546905
Skewness	-0.781445
Excess kurtosis	3.979439
Jarque-Bera stat.	3013.172
Jarque-Bera p-value	0

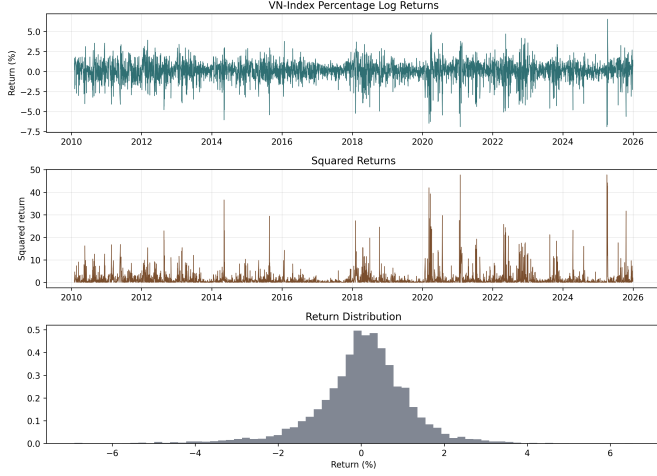


Fig. 1. VN-Index returns, squared returns, and return distribution. The time series shows volatility clustering and occasional large shocks.

date $t + 1$. Returns are measured in percentage units, so v_t and y_t are percentage-squared variance proxies.

Squared daily return is observable and simple, but noisy. A single close-to-close return can be small even when intraday variation is high, and a single jump can dominate the proxy. For this reason, the paper uses QLIKE as the main loss and reports robustness checks with daily OHLC range-based volatility proxies.

C. Descriptive Return Evidence

Table II shows that VN-Index daily returns are not approximately Gaussian. The mean return is small relative to the standard deviation, while skewness is negative and excess kurtosis is high. The Jarque-Bera statistic strongly rejects normality. This matters for volatility and VaR because extreme observations are more frequent than a Normal model would suggest, and downside shocks are economically important for market-risk applications.

D. Volatility Clustering and ARCH Effects

Table III provides formal motivation for conditional-variance modeling. Ljung-Box tests on squared returns and ARCH-LM tests reject the absence of volatility dependence at lags 5, 10, and 20. Raw return dependence is weaker and economically less central than squared-return dependence. This is the standard econometric pattern behind ARCH and

TABLE III
RETURN-DEPENDENCE AND ARCH-EFFECT DIAGNOSTICS.

Test	Series	Lag	Statistic	p-value
Ljung-Box	returns	5	41.839	6.35e-08
Ljung-Box	returns	10	49.343	3.52e-07
Ljung-Box	returns	20	60.711	5.53e-06
Ljung-Box	squared_returns	5	794.846	1.51e-169
Ljung-Box	squared_returns	10	994.918	2.33e-207
Ljung-Box	squared_returns	20	1081.398	1.67e-216
ARCH-LM	demeaned_returns	5	455.264	3.6e-96
ARCH-LM	demeaned_returns	10	480.760	5.69e-97
ARCH-LM	demeaned_returns	20	496.552	1.53e-92

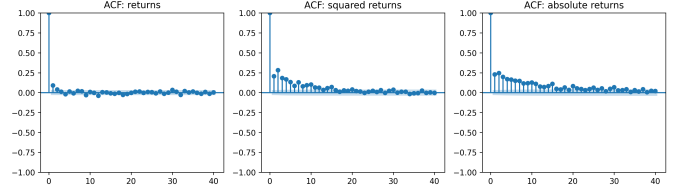


Fig. 2. ACF diagnostics for returns, squared returns, and absolute returns. Squared and absolute returns display more persistent dependence than raw returns.

GARCH models: conditional means are hard to forecast, but conditional variance is persistent.

E. OHLC Volatility Proxies

The daily proxy robustness analysis uses close-to-close squared return, Parkinson, Garman-Klass, and Rogers-Satchell variance proxies as same-horizon daily alternatives. Rolling Yang-Zhang 5-, 10-, and 20-day proxies are retained only as smoothed sensitivity checks because they are overlapping measurements and not identical to the one-day-ahead squared-return target. The main paper therefore emphasizes the daily proxy results and treats rolling proxy rankings as appendix-level sensitivity evidence.

IV. FORECASTING MODELS AND EXPERIMENTAL PROTOCOL

A. Benchmark Models

The benchmark set begins with simple variance forecasts. The historical mean predicts the training-sample average squared return. RollingVol-5, RollingVol-10, and RollingVol-20 use trailing rolling volatility, squared to produce variance forecasts. These baselines are intentionally simple, but they are useful because volatility persistence can make recent realized variation competitive.

The EWMA/RiskMetrics model follows

$$\sigma_{t+1}^2 = \lambda \sigma_t^2 + (1 - \lambda) r_t^2, \quad (4)$$

where r_t^2 is the current squared percentage return. The decay parameter is selected by validation QLIKE from $\lambda \in \{0.90, 0.94, 0.97, 0.99\}$, and the selected model is EWMA($\lambda = 0.90$).

HAR-type models approximate heterogeneous volatility persistence with daily, weekly, and monthly components:

$$y_{t+1} = \beta_0 + \beta_d RV_{d,t} + \beta_w RV_{w,t} + \beta_m RV_{m,t} + \varepsilon_{t+1}. \quad (5)$$

TABLE IV
ADVANCED GARCH FAILURE TAXONOMY. CATEGORIES SUM EXACTLY
TO THE 576 FAILED CANDIDATES.

Failure category	Count	Example
invalid_forecast_start_for_ar_mean_lag	432	AR(2), ARCH(1), normal
nonfinite_forecast	144	AR(1), ARCH(1), normal

HAR-SquaredReturn uses squared-return components. HAR-Parkinson replaces them with daily and averaged Parkinson range-variance components and is selected as the primary HAR variant by validation QLIKE.

B. GARCH-Family Models

The standard GARCH(1,1) benchmark is

$$r_t = \mu + \epsilon_t, \quad \epsilon_t = \sigma_t z_t, \quad (6)$$

$$\sigma_t^2 = \omega + \alpha \epsilon_{t-1}^2 + \beta \sigma_{t-1}^2. \quad (7)$$

Parameters are estimated on the training split for the static protocol. Validation and test forecasts use fixed estimated parameters while recursively updating the volatility state with returns observed before each forecast origin.

ARIMA-GARCH first models return mean dynamics using a training-AIC ARIMA selection over $p, q \in \{0, 1, 2, 3\}$ with $d = 0$, then fits GARCH(1,1) to the residuals. The original benchmark selects ARIMA(0,0,3).

The advanced GARCH-family search evaluates direct ARCH-family and two-step ARIMA-GARCH candidates. It includes ARCH, GARCH, EGARCH, GJR-GARCH, APARCH, and FIGARCH-type specifications with Normal, Student- t , skewed- t , and GED innovations where supported. Candidate selection uses validation QLIKE only. The primary advanced model is AdvGARCH-BestQLIKE, an ARIMA(3,0,3)-EGARCH(1,1,2)-Normal specification. AdvGARCH-BestAsymmetric is reported as a category-selected asymmetric comparator, not as the primary winner merely because its test QLIKE is slightly lower.

The search attempts 3,168 candidates, with 2,592 successful fits and 576 failures. Table IV reconciles the failures: 432 are invalid forecast-start failures for AR-mean lag specifications and 144 are non-finite forecast failures. This audit is important because large specification searches can look precise while hiding numerical fragility.

C. Neural, Hybrid, Calibration, and Combination Models

The base LSTM uses length-20 sequences of lagged market and volatility features, including returns, squared returns, absolute returns, rolling volatility, volume, and trading value. The ARIMA-GARCH-LSTM hybrid augments these inputs with econometric variance and residual-related features. Tuned neural variants include log-target training and QLIKE-style loss training; selection is by validation QLIKE within the neural group.

Calibration methods include multiplicative QLIKE scaling, mean matching, nonnegative affine calibration, and isotonic calibration fitted on validation data only. Forecast combinations include equal-weight means, medians, two-model convex

combinations, and stacking weights selected by validation QLIKE. Test-best combinations are treated as exploratory and are not used as primary evidence.

The neural sequence-construction code has been corrected so validation and test forecasts may use valid pre-split historical context without target leakage. The checked-in neural forecasts were not retrained in the final pass, so the primary comparison remains the fair 726-date common window shared by all existing primary artifacts.

D. Evaluation Protocol

For aligned observations $\{y_t, \hat{y}_t\}_{t=1}^n$, the evaluation uses

$$\text{RMSE} = \sqrt{\frac{1}{n} \sum_{t=1}^n (y_t - \hat{y}_t)^2}, \quad (8)$$

$$\text{MAE} = \frac{1}{n} \sum_{t=1}^n |y_t - \hat{y}_t|, \quad (9)$$

$$\text{QLIKE} = \frac{1}{n} \sum_{t=1}^n \left(\log(\hat{y}_t) + \frac{y_t}{\hat{y}_t} \right). \quad (10)$$

Predicted variances are clipped at $\epsilon = 10^{-8}$ before QLIKE computation. QLIKE is central because severe variance underprediction is penalized through $y_t/\hat{y}_t$.

Paired comparisons align observations by origin and target date, not by row order. Diebold-Mariano-style tests use loss differences relative to GARCH(1,1), and Holm correction is applied within clearly defined loss families. Regime and spike diagnostics use the common-window actual variance distribution. VaR diagnostics use a common Normal variance-to-VaR mapping so that all variance forecasts are translated into risk signals under the same distributional assumption. This is not a complete model-specific tail-distribution evaluation.

E. Expanding-Window Refit Robustness

The refit robustness experiment does not rerun the full advanced search. It fixes the validation-selected specifications and compares the static train-only forecast protocol with an expanding-window monthly refit protocol for GARCH(1,1), ARIMA-GARCH, EWMA($\lambda = 0.90$), HAR-Parkinson, AdvGARCH-BestQLIKE, and AdvGARCH-BestAsymmetric. At each monthly refit date, parameters are estimated using observations available before that month, and one-step forecasts are generated for that month. Failed refits are recorded explicitly rather than silently dropped.

V. EMPIRICAL RESULTS

A. Primary Common-Window Leaderboard

Table V is the authoritative primary leaderboard. Every listed model is evaluated on the same 726 target dates from 2023-02-07 to 2025-12-31 and the same next-day squared percentage log-return target. This resolves the earlier artifact mismatch between available-window econometric outputs and shorter neural sequence outputs.

TABLE V
PRIMARY COMMON-WINDOW TEST RESULTS. ALL MODELS USE TARGET DATES 2023-02-07 TO 2025-12-31 AND THE SAME NEXT-DAY SQUARED PERCENTAGE LOG-RETURN TARGET.

Rank	Model	Selection role	N	QLIKE	RMSE	MAE	Pred./Actual
1	AdvGARCH-BestAsymmetric	validation_selected_asymmetric_garch_comparator	726	1.021413	3.565782	1.314342	0.933
2	AdvGARCH-BestQLIKE	validation_selected_primary_advanced_garch	726	1.022898	3.583987	1.318196	0.931
3	HAR-Parkinson	validation_selected_har	726	1.029373	3.621325	1.353338	0.971
4	HAR-SquaredReturn	predeclared_har_comparator	726	1.039364	3.590457	1.355566	0.993
5	GARCH(1,1)	training_selected_econometric_benchmark	726	1.052888	3.648839	1.393909	1.024
6	ARIMA-GARCH	training_aic_mean_model_plus_garch_benchmark	726	1.055679	3.671816	1.400256	1.020
7	ComboStack-PoolD_AllStable	validation_selected_forecast_combination	726	1.113530	3.665685	1.671985	1.398
8	Calibrated-Hybrid-LogTarget-Small-isotonic	validation_selected_calibrated_neural	726	1.118671	3.666914	1.619575	1.326
9	LSTM	validation_monitored_neural_baseline	726	1.161001	3.769716	1.639734	1.294
10	ARIMA-GARCH-LSTM	validation_monitored_hybrid_neural_baseline	726	1.168609	3.763253	1.720113	1.404
11	HistoricalMean	fixed_baseline	726	1.192259	3.799706	1.449153	0.989
12	EWMA(lambda=0.90)	validation_selected_ewma	726	1.219608	3.716363	1.410256	1.000
13	Hybrid-QLIKE	validation_selected_tuned_hybrid_neural	726	1.230508	3.807296	1.862578	1.549
14	RollingVol-20	fixed_baseline	726	1.280635	3.901051	1.462196	0.997
15	RollingVol-10	fixed_baseline	726	1.309964	3.974129	1.478891	1.005
16	RollingVol-5	fixed_baseline	726	1.928009	4.011383	1.492394	0.992

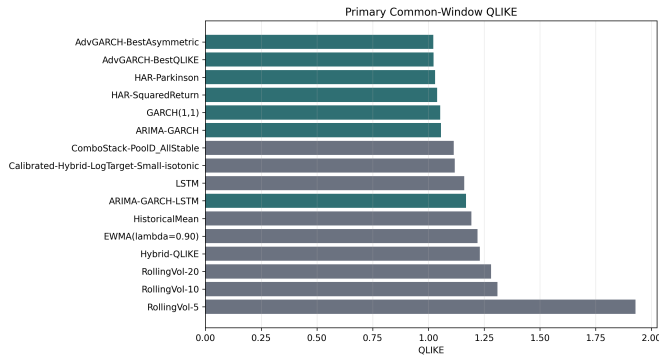


Fig. 3. Primary common-window QLIKE for the verified 726-observation leaderboard. Lower is better.

The strongest group consists of selected GARCH-family specifications and HAR-Parkinson. AdvGARCH-BestAsymmetric has the lowest common-window QLIKE, 1.021413, but it is a category-selected asymmetric comparator. The primary validation-selected advanced model, AdvGARCH-BestQLIKE, is very close at 1.022898. HAR-Parkinson follows with QLIKE 1.029373, ahead of HAR-SquaredReturn at 1.039364, GARCH(1,1) at 1.052888, and ARIMA-GARCH at 1.055679.

The neural and hybrid models do not robustly improve on these econometric benchmarks. The base LSTM has QLIKE 1.161001 and the ARIMA-GARCH-LSTM has QLIKE 1.168609. The best validation-selected calibrated neural forecast in the primary inventory improves to 1.118671, but remains behind the strongest GARCH-family and HAR forecasts. This is not evidence that neural volatility models generally fail; it shows that the implemented neural and hybrid specifications do not dominate in this daily VN-Index setting.

B. Parameter and Residual Diagnostics

Table VI gives the post-estimation financial econometric diagnostics. For GARCH(1,1), $\alpha = 0.1257$, $\beta = 0.8476$, and $\alpha + \beta = 0.9733$. The persistence is below one, so the unconditional variance and shock half-life are technically

defined; the estimated half-life is about 25.6 trading days. This supports the economic interpretation that volatility shocks to the VN-Index decay slowly.

The two EGARCH specifications also display high log-variance persistence, with beta-sum measures near 0.95. Both AdvGARCH-BestQLIKE and AdvGARCH-BestAsymmetric have negative and statistically significant $\gamma[1]$ estimates at conventional levels. Under the EGARCH sign convention, this is consistent with negative shocks raising future volatility more than positive shocks of comparable standardized magnitude. The evidence supports an asymmetric volatility response, but it should still be read cautiously because the models are selected from a broad specification search.

Table VII shows that squared standardized residual dependence is largely removed for the selected GARCH-family models. Ljung-Box tests on squared standardized residuals and ARCH-LM tests do not reject at lags 5, 10, or 20. This contrasts with the strong ARCH-LM rejections in raw returns and indicates that the models capture much of the conditional heteroskedasticity. Standardized residuals remain non-normal, however, so heavy tails and tail-risk calibration remain important.

C. Statistical Comparisons

The common-window Diebold-Mariano outputs are intentionally interpreted conservatively. Against GARCH(1,1), AdvGARCH-BestAsymmetric has a raw QLIKE comparison $p = 0.0015$ and Holm-adjusted $p = 0.0215$ within the final primary QLIKE family. AdvGARCH-BestQLIKE has raw $p = 0.0181$, but its Holm-adjusted $p = 0.2354$ does not support a decisive superiority claim after family-wise correction. HAR-Parkinson is close to GARCH(1,1), but the paired QLIKE difference is not statistically decisive. The statistical message is therefore class-level rather than single-winner: selected GARCH-family specifications and HAR-Parkinson are the strongest group.

TABLE VI
SELECTED GARCH-FAMILY PARAMETER DIAGNOSTICS. PERSISTENCE AND HALF-LIFE ARE REPORTED ONLY WHERE THE FORMULA IS TECHNICALLY APPROPRIATE.

Model	Variance equation	Parameter	Estimate	Std. err.	<i>p</i> -value	Persistence	Validity
GARCH(1,1)	GARCH(1,1)	mu	0.0456	0.0204	0.0256	0.9733	stationary
GARCH(1,1)	GARCH(1,1)	omega	0.0374	0.0163	0.022	0.9733	stationary
GARCH(1,1)	GARCH(1,1)	alpha[1]	0.1257	0.0212	3.16e-09	0.9733	stationary
GARCH(1,1)	GARCH(1,1)	beta[1]	0.8476	0.0303	1.61e-172	0.9733	stationary
AdvGARCH-BestQLIKE	EGARCH(1,1,2)	omega	0.0063	0.0057	0.273	0.9503	EGARCH persist.
AdvGARCH-BestQLIKE	EGARCH(1,1,2)	alpha[1]	0.2380	0.0271	1.84e-18	0.9503	EGARCH persist.
AdvGARCH-BestQLIKE	EGARCH(1,1,2)	gamma[1]	-0.0440	0.0186	0.0178	0.9503	EGARCH persist.
AdvGARCH-BestQLIKE	EGARCH(1,1,2)	beta[1]	0.9503	0.1492	1.91e-10	0.9503	EGARCH persist.
AdvGARCH-BestQLIKE	EGARCH(1,1,2)	beta[2]	0.0000	0.1536	1	0.9503	EGARCH persist.
AdvGARCH-BestAsymmetric	EGARCH(1,1,1)	omega	0.0088	0.0056	0.116	0.9532	EGARCH persist.
AdvGARCH-BestAsymmetric	EGARCH(1,1,1)	alpha[1]	0.2353	0.0283	1e-16	0.9532	EGARCH persist.
AdvGARCH-BestAsymmetric	EGARCH(1,1,1)	gamma[1]	-0.0430	0.0189	0.0228	0.9532	EGARCH persist.
AdvGARCH-BestAsymmetric	EGARCH(1,1,1)	beta[1]	0.9532	0.0139	0	0.9532	EGARCH persist.

TABLE VII
POST-FIT RESIDUAL DIAGNOSTICS FOR SELECTED GARCH-FAMILY MODELS.

Model	Test	Series	Lag	Statistic	<i>p</i> -value	Flag
GARCH(1,1)	Ljung-Box	Squared std. resid.	5	1.283	0.937	not_rejected
GARCH(1,1)	Ljung-Box	Squared std. resid.	10	3.417	0.97	not_rejected
GARCH(1,1)	Ljung-Box	Squared std. resid.	20	14.273	0.816	not_rejected
GARCH(1,1)	ARCH-LM	Std. resid.	5	1.267	0.938	not_rejected
GARCH(1,1)	ARCH-LM	Std. resid.	10	3.357	0.972	not_rejected
GARCH(1,1)	ARCH-LM	Std. resid.	20	13.936	0.834	not_rejected
AdvGARCH-BestQLIKE	Ljung-Box	Squared std. resid.	5	2.309	0.805	not_rejected
AdvGARCH-BestQLIKE	Ljung-Box	Squared std. resid.	10	3.789	0.956	not_rejected
AdvGARCH-BestQLIKE	Ljung-Box	Squared std. resid.	20	15.057	0.773	not_rejected
AdvGARCH-BestQLIKE	ARCH-LM	Std. resid.	5	2.273	0.81	not_rejected
AdvGARCH-BestQLIKE	ARCH-LM	Std. resid.	10	3.699	0.96	not_rejected
AdvGARCH-BestQLIKE	ARCH-LM	Std. resid.	20	14.517	0.803	not_rejected
AdvGARCH-BestAsymmetric	Ljung-Box	Squared std. resid.	5	1.938	0.858	not_rejected
AdvGARCH-BestAsymmetric	Ljung-Box	Squared std. resid.	10	3.249	0.975	not_rejected
AdvGARCH-BestAsymmetric	Ljung-Box	Squared std. resid.	20	15.089	0.771	not_rejected
AdvGARCH-BestAsymmetric	ARCH-LM	Std. resid.	5	1.921	0.86	not_rejected
AdvGARCH-BestAsymmetric	ARCH-LM	Std. resid.	10	3.213	0.976	not_rejected
AdvGARCH-BestAsymmetric	ARCH-LM	Std. resid.	20	14.528	0.803	not_rejected

TABLE VIII
STATIC VERSUS EXPANDING-WINDOW MONTHLY REFIT ROBUSTNESS ON THE UNIFIED COMMON WINDOW.

Model	Protocol	<i>N</i>	Full	QLIKE	RMSE	MAE	Pred./Actual	Q90 QLIKE	Failed
ARIMA-GARCH	expanding_monthly	726	yes	1.049005	3.669563	1.417771	1.049	6.521	0
ARIMA-GARCH	static_train_only	726	yes	1.055679	3.671816	1.400256	1.020	6.770	0
AdvGARCH-BestAsymmetric	expanding_monthly	726	yes	1.014849	3.556510	1.350518	0.997	6.388	0
AdvGARCH-BestAsymmetric	static_train_only	726	yes	1.021413	3.565782	1.314342	0.933	6.713	0
AdvGARCH-BestQLIKE	expanding_monthly	340	no	1.202280	4.350815	1.532191	0.927	7.762	19
AdvGARCH-BestQLIKE	static_train_only	726	yes	1.022898	3.583987	1.318196	0.931	6.730	0
EWMA(lambda=0.90)	expanding_monthly	726	yes	1.219608	3.716363	1.410256	1.000	8.425	0
EWMA(lambda=0.90)	static_train_only	726	yes	1.219608	3.716363	1.410256	1.000	8.425	0
GARCH(1,1)	expanding_monthly	726	yes	1.047945	3.656469	1.415344	1.054	6.523	0
GARCH(1,1)	static_train_only	726	yes	1.052888	3.648839	1.393909	1.024	6.766	0
HAR-Parkinson	expanding_monthly	726	yes	1.029873	3.647880	1.374900	1.007	6.130	0
HAR-Parkinson	static_train_only	726	yes	1.029373	3.621325	1.353338	0.971	6.337	0

D. Rolling-Origin Refit Robustness

Table VIII compares static train-only forecasts with expanding-window monthly refits for the small preselected econometric set. The conclusion largely survives. AdvGARCH-BestAsymmetric remains the best complete-window refit model, improving from static QLIKE 1.021413 to expanding-monthly QLIKE 1.014849. GARCH(1,1) improves from 1.052888 to 1.047945, and ARIMA-GARCH improves from 1.055679 to 1.049005. HAR-Parkinson remains stable, with essentially unchanged QLIKE around 1.03.

The robustness experiment also reveals operational fragility.

AdvGARCH-BestQLIKE is the primary validation-selected advanced specification in the static protocol, but its monthly expanding refit fails in 19 monthly segments and only produces 340 common-window forecasts. That incomplete refit has QLIKE 1.202280 and is marked as not covering the full common window. This does not invalidate the static primary result, but it weakens the practical attractiveness of using that ARIMA-EGARCH specification in a monthly operational refit setting.

Periodic updating improves some complete econometric forecasts and reduces the one-percent common-Normal VaR violation count for several models. For example, GARCH(1,1)

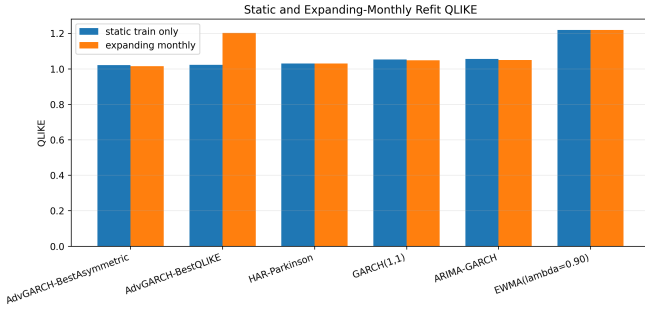


Fig. 4. Static and expanding-monthly QLIKE for the controlled refit robustness experiment.

declines from 15 to 14 one-percent violations after monthly refitting, while AdvGARCH-BestAsymmetric declines from 14 to 13. The changes are useful but do not remove the need for tail-risk diagnostics.

E. Spike and VaR Diagnostics

Average QLIKE hides a common risk weakness. Table IX evaluates days above the 90th percentile of actual variance. The strongest econometric models still underpredict most spike days, and severe underprediction rates remain high. HAR-Parkinson has the best spike-day QLIKE among the primary econometric models in the table, but it still severely underpredicts about three quarters of spike days. Spike diagnostics therefore qualify rather than overturn the average-loss ranking.

Table X maps variance forecasts to VaR using a common Normal quantile. At the 5% level, the main econometric models have violation counts near expectation. At the 1% level, strict-tail coverage is weaker. GARCH(1,1) has 15 violations out of 726 observations against 7.26 expected, with Kupiec $p = 0.0116$. AdvGARCH-BestQLIKE and AdvGARCH-BestAsymmetric each have 14 violations, with Kupiec $p = 0.0258$. These results show that good average volatility forecasts do not automatically deliver adequate extreme-risk calibration under the common-Normal mapping.

This is a standardized risk translation, not a full distribution-specific VaR evaluation. Heavy-tailed or skewed GARCH models may require model-specific quantiles, but the current primary prediction artifacts do not retain all fitted tail-shape and skew parameters needed for a clean distribution-specific comparison.

F. Proxy Robustness

The proxy robustness exercise confirms that exact rankings are sensitive to how latent volatility is observed. Daily close-to-close, Parkinson, Garman-Klass, and Rogers-Satchell proxies measure related but non-identical volatility objects. The main conclusion remains class-level: GARCH-family specifications are consistently competitive, and HAR-Parkinson is strong on the squared-return target because it uses OHLC range information directly. Rolling Yang-Zhang sensitivities are kept separate because they are smoothed, overlapping measurements rather than direct one-day targets.

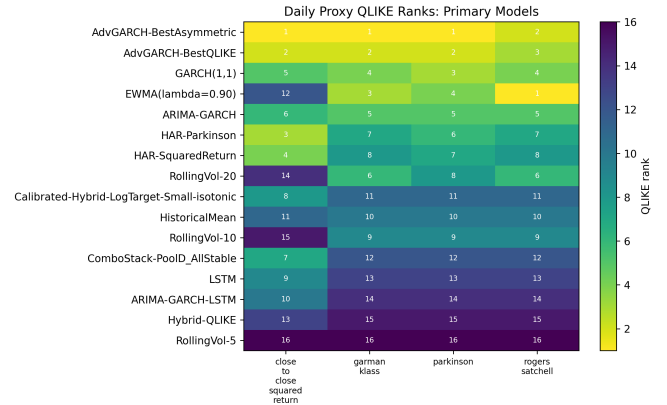


Fig. 5. Daily same-horizon proxy robustness for primary models. Numbers are QLIKE ranks; lower ranks are better.

G. Discussion

The empirical findings are economically coherent. Raw VN-Index returns are heavy-tailed and negatively skewed, while squared returns exhibit strong dependence. GARCH-family models perform well because they encode this persistence directly. HAR-Parkinson performs competitively because daily range information carries volatility information that close-to-close squared returns alone may miss.

The neural and hybrid findings are best interpreted as modest and dataset-specific. Daily sample size is limited for deep sequence models, and smooth neural forecasts can look acceptable under some average losses while missing spike timing or tail risk. Calibration and combinations can improve scale, but the implemented neural and hybrid approaches do not provide robust superiority over the strongest econometric benchmarks.

VI. LIMITATIONS

The study uses daily data only and lacks high-frequency realized volatility. Squared returns and OHLC range estimators are imperfect proxies for latent volatility, so proxy robustness is informative but not a substitute for realized-variance targets.

The advanced GARCH search is broad and numerically fragile. Although the static validation-selected models are useful, the refit experiment shows that AdvGARCH-BestQLIKE is not operationally stable under monthly expanding refits. The asymmetric comparator performs well, but it remains a category-selected robustness model rather than the sole primary winner.

Distribution-specific VaR is not implemented in the final evidence because the prediction artifacts do not cleanly retain all innovation distribution parameters. Neural sequence construction has been corrected in code, but full retraining into versioned output paths was not performed in the final pass. This does not affect the fairness of the primary leaderboard, because all primary models are compared on the same 726 target dates.

TABLE IX
COMMON-WINDOW SPIKE-DAY DIAGNOSTICS AT THE 90TH PERCENTILE OF ACTUAL VARIANCE.

Model	Spike days	Underpred.	Severe	Spike QLIKE
AdvGARCH-BestAsymmetric	73	97.3%	76.7%	6.713
AdvGARCH-BestQLIKE	73	94.5%	75.3%	6.730
HAR-Parkinson	73	93.2%	76.7%	6.337
GARCH(1,1)	73	94.5%	72.6%	6.766
ARIMA-GARCH	73	94.5%	74.0%	6.770
EWMA(lambda=0.90)	73	93.2%	72.6%	8.425
LSTM	73	100.0%	56.2%	5.099
ARIMA-GARCH-LSTM	73	100.0%	56.2%	4.793

TABLE X
COMMON-WINDOW COMMON-NORMAL VAR BACKTESTS.

Model	Level	N	Viol.	Expected	Kupiec p
AdvGARCH-BestAsymmetric	5%	726	33	36.30	0.568
AdvGARCH-BestAsymmetric	1%	726	14	7.26	0.0258
AdvGARCH-BestQLIKE	5%	726	33	36.30	0.568
AdvGARCH-BestQLIKE	1%	726	14	7.26	0.0258
HAR-Parkinson	5%	726	32	36.30	0.455
HAR-Parkinson	1%	726	15	7.26	0.0116
GARCH(1,1)	5%	726	34	36.30	0.692
GARCH(1,1)	1%	726	15	7.26	0.0116
ARIMA-GARCH	5%	726	34	36.30	0.692
ARIMA-GARCH	1%	726	15	7.26	0.0116
EWMA(lambda=0.90)	5%	726	37	36.30	0.905
EWMA(lambda=0.90)	1%	726	17	7.26	0.00197

VII. CONCLUSION

This study provides a reproducible financial econometric evaluation of one-step-ahead daily VN-Index volatility forecasts. Under a leakage-safe unified common-window protocol, selected GARCH-family specifications and HAR-Parkinson form the strongest group of forecasts. The best common-window QLIKE values are 1.021413 for the category-selected AdvGARCH-BestAsymmetric model, 1.022898 for the primary validation-selected AdvGARCH-BestQLIKE model, and 1.029373 for HAR-Parkinson.

The results are economically consistent with significant volatility clustering and persistent conditional variance in Vietnamese equity-index returns. The competitive performance of HAR-Parkinson indicates that daily OHLC range information is useful when high-frequency realized-volatility data are unavailable. The implemented neural and hybrid models do not robustly outperform the strongest econometric benchmarks in this daily-data setting, although calibrated and combined forecasts may still contain complementary information.

Risk diagnostics temper the average-loss findings. Spike underprediction remains common, and the 1% common-Normal VaR mapping rejects unconditional coverage for several strong average-loss models. The main lesson is therefore not that one model universally dominates. A credible daily VN-Index volatility forecast should be judged jointly by validation-based average loss, proxy robustness, operational refit stability, spike behavior, and tail-risk calibration.

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